

WASP-10b

R-1064

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_151128 · created 2026-07-30T12:41:20+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457354.7117 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1909 +/- 0.0095
Transit depth (Rp/Rs)^2	3.64 +/- 0.36 [%]
Orbital Inclination (inc)	88.34 +/- 0.71
Airmass coefficient 1 (a1)	1.037 +/- 0.014
Airmass coefficient 2 (a2)	-0.0235 +/- 0.0065
Scatter in the residuals of the lightcurve fit is	1.32 %
Best Comparison Star	#1 - [281, 128]
Optimal Method	PSF photometry
Transit Duration (day)	0.0934 +/- 0.0035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1909 $\pm$ 0.0095 vs 0.1592 (deviation 3.34 σ)
Duration fit vs published	0.0934 d vs 0.0946 d (deviation 0.34 σ)
Mid-time O-C	0.2 min
Depth SNR	20.1
Residual scatter	1.32 %

Light curve

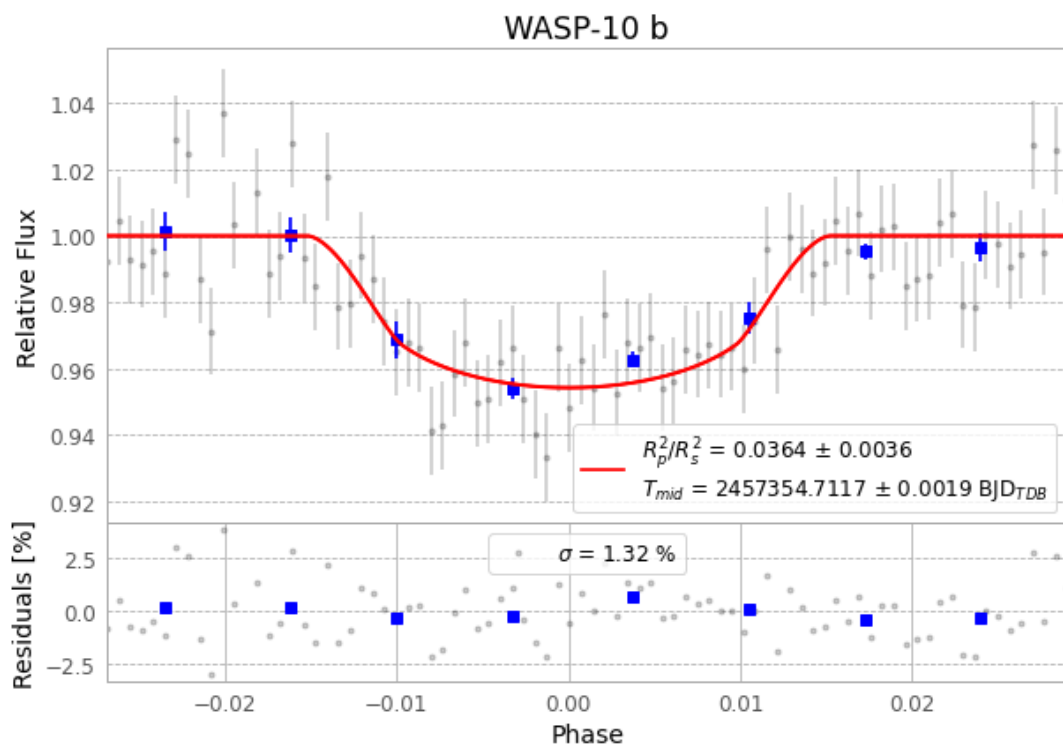


Figure 1 — FinalLightCurve_WASP-10 b_2015-11-27.png (SHA-256 301ad7f91243dfb8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [285, 226], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 8ff169e1f2381ccc...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

84 FITS frames (55 MB), WASP-10151128030012.FITS → WASP-10151128070912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-151128.log (05fb45b156d8...), FinalLightCurve_WASP-10 b_2015-11-27.png (301ad7f91243...), FinalLightCurve_WASP-10 b_2015-11-27.csv (78f43ebd54cf...)

Compute resources

Wall time 115.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 12:41Z.